

Sample Paper 1 — Class 5 Mathematics Olympiad

Pattern: SOF IMO Level 1 — 50 questions, 60 marks, 60 minutes, no negative marking. Set a timer before you start (or do it untimed the first time, per Week 11 of the practice plan).

Section 1 — Logical Reasoning (Q1–Q15, 1 mark each)

- Q1.** Find the odd one out: A) Square B) Rectangle C) Circle D) Triangle
- Q2.** Rose is to Flower as Lion is to _____. A) Forest B) Animal C) Mane D) Roar
- Q3.** If each letter of a word is moved one step forward in the alphabet ($A \rightarrow B, B \rightarrow C, \dots$), DOG becomes EPH. Using the same rule, what does CAT become? A) DBU B) DCU C) DBV D) CBU
- Q4.** Complete the series: 4, 8, 13, 19, 26, ____ A) 32 B) 33 C) 34 D) 35
- Q5.** In a row of 20 students, Meena is 8th from the left. What is her position from the right? A) 12th B) 13th C) 14th D) 11th
- Q6.** Facing North, Rohit turns 90° clockwise, then another 90° clockwise. Which direction does he face now? A) East B) South C) West D) North
- Q7.** Arrange in a logical growth order: 1. Seed 2. Tree 3. Flower 4. Fruit 5. Sapling A) 1, 5, 2, 3, 4 B) 1, 2, 5, 3, 4 C) 1, 5, 3, 2, 4 D) 5, 1, 2, 3, 4
- Q8.** Find the odd one out: A) 17 B) 19 C) 21 D) 23
- Q9.** If today is Wednesday, what day will it be after 17 days? A) Friday B) Saturday C) Sunday D) Thursday
- Q10.** Pointing to a man, Sita said, "He is the son of my mother's only brother." How is the man related to Sita? A) Brother B) Cousin C) Nephew D) Uncle
- Q11.** Find the missing number: 2, 5, 11, 23, ____ A) 43 B) 45 C) 47 D) 49
- Q12.** In a certain code, MANGO is written as OCPIQ (each letter moved 2 steps forward). How is TABLE written in the same code? A) VCDNG B) VBDNG C) VCDMG D) WCDNG
- Q13.** Find the odd one out: A) Cube B) Sphere C) Cone D) Square
- Q14.** A is taller than B. C is shorter than B. D is taller than A. Who is the tallest? A) A B) B C) C D) D
- Q15.** Look at the pattern: 1, 4, 9, 16, 25, ____ A) 30 B) 32 C) 36 D) 49

Section 2 — Mathematical Reasoning (Q16–Q35, 1 mark each)

- Q16.** $45,678 + 23,459 = ?$ A) 69,137 B) 69,127 C) 68,137 D) 69,237
- Q17.** $80,005 - 34,567 = ?$ A) 45,438 B) 45,338 C) 45,448 D) 46,438
- Q18.** $3,456 \times 8 = ?$ A) 27,648 B) 27,548 C) 27,748 D) 27,638
- Q19.** $9,576 \div 7 = ?$ A) 1,368 B) 1,358 C) 1,378 D) 1,468
- Q20.** Find the HCF of 24 and 36. A) 6 B) 8 C) 12 D) 18
- Q21.** Find the LCM of 6 and 8. A) 24 B) 48 C) 12 D) 16
- Q22.** Which fraction is equivalent to $\frac{3}{4}$? A) $\frac{6}{8}$ B) $\frac{4}{3}$ C) $\frac{3}{8}$ D) $\frac{6}{9}$
- Q23.** $\frac{2}{5} + \frac{1}{5} + \frac{1}{5} = ?$ A) $\frac{4}{5}$ B) $\frac{4}{15}$ C) $\frac{3}{5}$ D) 1
- Q24.** Convert 0.75 to a fraction in simplest form. A) $\frac{3}{4}$ B) $\frac{7}{5}$ C) $\frac{3}{5}$ D) $\frac{7}{10}$
- Q25.** What is 15% of 200? A) 20 B) 25 C) 30 D) 35
- Q26.** What is $\frac{3}{4}$ expressed as a percentage? A) 34% B) 43% C) 75% D) 134%
- Q27.** The ratio of pens to pencils is 2:3. If there are 12 pens, how many pencils are there? A) 15 B) 16 C) 18 D) 20
- Q28.** A shopkeeper buys a toy for ₹80 and sells it for ₹100. Find the profit percentage. A) 20% B) 25% C) 30% D) 15%
- Q29.** Find the simple interest on ₹1,000 at 5% per annum for 2 years. A) ₹50 B) ₹100 C) ₹150 D) ₹200
- Q30.** Convert 4 kg 250 g into grams. A) 4250 g B) 4025 g C) 4205 g D) 42500 g
- Q31.** Convert 3.5 hours into minutes. A) 180 min B) 200 min C) 210 min D) 215 min
- Q32.** Find the perimeter of a rectangle with length 14 cm and breadth 9 cm. A) 23 cm B) 46 cm C) 126 cm D) 92 cm
- Q33.** Find the area of a square with side 12 cm. A) 48 sq cm B) 120 sq cm C) 144 sq cm D) 164 sq cm
- Q34.** How many lines of symmetry does a rectangle (non-square) have? A) 1 B) 2 C) 4 D) 0
- Q35.** How many right angles are there in a full turn around a point? A) 2 B) 3 C) 4 D) 6

Section 3 — Everyday Mathematics (Q36–Q45, 1 mark each)

- Q36.** A factory produces 2,350 bottles each day. How many bottles does it produce in 30 days? A) 70,500 B) 69,500 C) 71,500 D) 70,050
- Q37.** Riya had ₹5,000. She spent ₹1,250 on books and ₹875 on stationery. How much money does she have left? A) ₹2,875 B) ₹2,975 C) ₹2,785 D) ₹2,675

- Q38.** A tank holds 450 litres. If $\frac{1}{3}$ of it is used, how much water is left? A) 250 L B) 300 L C) 150 L D) 350 L
- Q39.** A shopkeeper bought 25 kg of rice for ₹1,250. What is the cost of 1 kg of rice? A) ₹40 B) ₹45 C) ₹50 D) ₹55
- Q40.** A school has 8 sections in Class 5, each with 42 students. How many students are there in Class 5 in total? A) 326 B) 336 C) 346 D) 356
- Q41.** Rakesh deposited ₹2,000 in a bank at 6% simple interest per year. How much interest will he earn in 3 years? A) ₹300 B) ₹320 C) ₹360 D) ₹400
- Q42.** A dealer bought a bicycle for ₹2,400 and sold it for ₹2,880. Find the profit percentage. A) 15% B) 18% C) 20% D) 25%
- Q43.** A train travels 60 km in 1 hour. How far will it travel in 2 hours 30 minutes at the same speed? A) 120 km B) 135 km C) 150 km D) 180 km
- Q44.** A recipe needs $\frac{3}{4}$ cup of sugar. How much sugar is needed for 4 such recipes? A) 2 cups B) 3 cups C) 4 cups D) 3.5 cups
- Q45.** In a survey, 60% of 150 students like cricket. How many students like cricket? A) 80 B) 85 C) 90 D) 95

Section 4 — Achievers Section (Q46–Q50, 3 marks each)

- Q46.** A number is increased by 20% to get 240. Find the original number. A) 180 B) 190 C) 200 D) 210
- Q47.** The sum of three consecutive even numbers is 72. Find the largest number. A) 22 B) 24 C) 26 D) 28
- Q48.** A sum of money amounts to ₹1,200 as simple interest in 4 years at 5% per annum. Find the principal. A) ₹5,000 B) ₹6,000 C) ₹7,000 D) ₹8,000
- Q49.** Meena had some money. She spent $\frac{1}{4}$ of it on a book and $\frac{1}{3}$ of the remainder on a pen. She is left with ₹600. How much money did she have at first? A) ₹1,000 B) ₹1,200 C) ₹1,400 D) ₹1,500
- Q50.** The ratio of ages of A and B is 3:5. After 4 years, the ratio becomes 4:6. Find A's present age. A) 9 B) 12 C) 15 D) 18

Answer key: `sample-paper-01-answer-key.md`
